Alokit Mishra

Machine Learning Engineer — Generative AI, Computer Vision & NLP

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Summary

Machine Learning Engineer with hands-on experience in **Generative AI, Computer Vision, NLP, OCR, RAG systems, and AI model deployment**. Skilled in building scalable AI solutions using **Python, PyTorch, Transformers, Redis, Docker, and Blockchain**. Experienced in developing production-ready ML pipelines, document verification systems, and intelligent analytics platforms with strong focus on scalability and performance optimization.

Education

Madan Mohan Malaviya University Of Technology — **Gorakhpur, UP**
*Bachelor of Technology in Computer Science, **CGPA:** 8.8/10* — *Aug 2024 – May 2028*

Projects

SatyaSetu: AI-Powered Document Verification Platform | *Node.js, Redis, Blockchain, YOLO, ViT, OCR* — **Dec 2025**

– Built scalable academic document verification platform for nationwide institutional deployment.
– Developed AI verification pipeline integrating **OCR**, **YOLO-based layout detection**, and **Vision Transformer (ViT)** models for forgery detection.
– Achieved **94% accuracy** in document tampering detection with support for **10,000+ concurrent verification requests**.
– Implemented backend infrastructure using **Redis caching**, asynchronous worker queues, and blockchain-backed verification logs. **Link:** satyasetu.live

GroqSense: Financial Intelligence Platform | *LLMs, RAG, Financial Analytics* — **Mar 2025**

– Built full-stack **RAG-based conversational platform** for real-time stock analysis and trend recognition.
– Implemented custom embedding models and vector database search for efficient financial document retrieval.
– Developed pattern recognition algorithms for market trend prediction and financial insight generation.

Structify AI: Autonomous Document Classification | *OpenCV, SVM, Computer Vision* — **Dec 2024**

– **Top 5, IIT Bombay Techfest** for developing AI-powered document classification system.
– Implemented advanced feature engineering techniques combining **HOG**, **SIFT**, and deep feature representations.
– Achieved **89% classification accuracy** across 12 document categories with inference time under **200ms**.

Experience

Machine Learning Engineer — **Remote, India**
Jax AI — *Present*

Operations Head — **MMMUT, Gorakhpur**
AI Spark Research Group — *Present*

Mentor – AI/ML & DSA Domain — **MMMUT, Gorakhpur**
Mentoring Junior Students in AI/ML & Data Structures and Algorithms — *Present*

Achievements

Smart India Hackathon (SIH) 2025 – Winner — **Dec 2025**
AI-Powered Document Verification Platform — *IIT (ISM) Dhanbad*

IIT Bombay Techfest – Top 5 — **Dec 2024**
Autonomous Document Classification System — *IIT Bombay*

Technical Skills

Languages: Python, C++, SQL, JavaScript
ML/AI: Generative AI, NLP, Computer Vision, CNNs, Transformers, LLMs, RAG, OCR, YOLOv8
Tools: PyTorch, OpenCV, Redis, Docker, Git, Linux, ONNX
Concepts: Model Evaluation, Feature Engineering, AI Deployment, REST APIs, Vector Databases

Relevant Coursework

Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Database Management Systems, Operating Systems, Computer Networks